

HTML & CSS

Complete Front-End

Roadmap

An 80-Day, Day-by-Day Plan for Job Readiness

Structure • Semantics • Styling • Flexbox • Grid • Responsive • Animation

8 Phases | 80 Days | 2 Capstone Projects

From first tag to deployed, responsive interfaces

How to Use This Roadmap

Spend one focused day on each topic below. Read and take notes in the morning, then immediately practice by writing real code — every concept here is something you should be able to demonstrate, not just recognise.

Days marked as “Project” or “Capstone” combine many topics; treat them as checkpoints. If a project day takes longer than one day, that is normal and worthwhile. The Final Job-Ready Checklist on the last pages is your interview-prep summary.

The 8 Phases

Days 1–18 Phase 1 | HTML Foundations

Days 19–22 Phase 2 | HTML — Accessibility, SEO & Practice

Days 23–40 Phase 3 | CSS Fundamentals

Days 41–54 Phase 4 | CSS Layout — Flexbox, Grid & Patterns

Days 55–62 Phase 5 | Responsive & Adaptive Design

Days 63–72 Phase 6 | Visual Effects, Transforms & Animation

Days 73–78 Phase 7 | Modern CSS, Architecture & Tooling

Days 79–80 Phase 8 | Capstone Projects & Job-Ready Review

Phase 1 • HTML Foundations

Day 1: Introduction to the Web & How It Works

How the Web Works

- Internet vs the World Wide Web
- Client-Server Model
- Request-Response Cycle (HTTP/HTTPS)
- What a Web Server Does
- URLs, Domains & DNS (overview)

How Browsers Render Pages

- Browser Rendering Engine Basics
- The Three Core Languages
 - HTML → Structure
 - CSS → Presentation
 - JavaScript → Behaviour
- Frontend vs Backend
- Web Standards (W3C / WHATWG)

Development Setup

- Installing a Code Editor (VS Code)
- Useful Extensions (Live Server, Prettier)
- Browser DevTools Overview
- Creating & Opening Your First .html File

Day 2: HTML Document Structure

Core Building Blocks

- What HTML Is (HyperText Markup Language)
- Elements, Tags & Attributes
 - Opening & Closing Tags
 - Attribute name="value" syntax
- Void / Self-Closing Elements
- Element Nesting Rules

The Document Skeleton

- `<!DOCTYPE html>` Declaration
- `<html>` Root Element
 - lang Attribute
 - dir Attribute
- `<head>` vs `<body>`
- Standard Boilerplate Structure

Good Habits

- Indentation & Readability
- Lowercase Tags & Quoted Attributes
- HTML Comments `<!-- -->`

Day 3: The Head Section & Metadata

Document Identity

- `<title>` Element
- `<meta charset="UTF-8">`
- `<meta name="viewport">` (responsive)
- Descriptive Meta Tags
 - description
 - keywords
 - author

Linking External Resources

- `<link rel="stylesheet">`
- `<link rel="icon">` (favicon)
- `<style>` Tag
- `<script>` Tag Placement
 - Head vs end of body
 - defer & async (intro)
- `<base>` Element

Day 4: Headings, Paragraphs & Text Blocks

Headings

- Heading Elements h1-h6
- Heading Hierarchy & Why It Matters
- One `<h1>` Per Page (best practice)

Paragraphs & Spacing

- `<p>` Paragraph Element
- Whitespace Collapsing
- Line Breaks `<br>`
- Horizontal Rule `<hr>`
- `<pre>` Preformatted Text

Day 5: Text Formatting & Inline Semantics

Importance & Emphasis

- Semantic vs Visual
 - `<strong>` vs `<b>`
 - `<em>` vs `<i>`
- `<mark>` Highlight
- `<small>`, `<del>`, `<ins>`
- `<sub>` & `<sup>`

Quotes & Technical Text

- `<blockquote>` & `<q>`
- `<cite>`, `<abbr>`
- Code-Related Tags
 - `<code>`
 - `<kbd>`
 - `<samp>`
 - `<var>`
- `<span>` Generic Inline Container

Day 6: HTML Entities & Symbols

Character Entities

- Why Entities Are Needed
- Reserved Characters
 - `<` & `>`
 - `&`
 - `"` & `'`
- Non-Breaking Space ` `

Symbols & Numbers

- Currency, Math & Arrow Symbols
- Named vs Numeric Entities
- Emoji in HTML

Day 7: Lists

List Types

- Unordered List
 - `<ul>`
 - `<li>`
- Ordered List
 - `<ol>`
 - type / start / reversed
- Description List
 - `<dl>`
 - `<dt>`
 - `<dd>`

Working With Lists

- Nested Lists
- Lists as Navigation Menus
- Choosing the Right List Type

Day 8: Hyperlinks (Anchors)

The Anchor Element

- `<a>` & the href Attribute
- URL Types
 - Absolute URLs
 - Relative URLs
 - Fragment / #id Links
- Special Links
 - `mailto:`
 - `tel:`

Link Behaviour & Safety

- target Attribute
 - `_self`
 - `_blank`
 - `_parent`
 - `_top`
- rel Attribute
 - `noopener`
 - `noreferrer`
 - `nofollow`
- download Attribute
- Link vs Button Semantics

Day 9: Images

The Image Element

- `<img>`, `src`, `alt`
- width & height Attributes
- Image Formats
 - JPG, PNG, GIF
 - WebP, AVIF
 - SVG (vector)
- Relative vs Absolute Paths

Modern Image Techniques

- `loading="lazy"`
- `<figure>` & `<figcaption>`
- Responsive Images
 - `srcset`
 - `sizes`
 - `<picture>` & art direction

Day 10: Tables — Structure

Building a Table

- Core Tags
 - `<table>`
 - `<tr>`
 - `<td>`
 - `<th>`
- Sectioning
 - `<thead>`
 - `<tbody>`
 - `<tfoot>`
- `<caption>`

Cell Control

- `colspan` & `rowspan`
- Header scope Attribute
- When to Use (and NOT use) Tables

Day 11: Tables — Advanced & Accessible

Columns & Grouping

- `<colgroup>` & `<col>`
- Styling Hooks for Columns

Accessible Data Tables

- headers / id Associations
- Complex Header Relationships
- Tabular Data Best Practices

Day 12: Forms — Introduction & Structure

The Form Element

- `<form>` Element
- Submission Basics
 - action Attribute
 - method: GET vs POST
- enctype Attribute

Labelling Inputs

- `<label>` & the for Attribute
- name & value Attributes
- Why Every Input Needs a Label

Day 13: Form Inputs — Text-Based

The Input Element

- `<input>` & the type Attribute
- Text Input Types
 - `text`
 - `password`
 - `email`
 - `url`
 - `tel`
 - `search`
- Numeric Inputs
 - `number`
 - `range`

Input Attributes

- placeholder
- State Attributes
 - `required`
 - `readonly`
 - `disabled`
- Constraints
 - `maxlength` / `minlength`
 - `pattern`
- `autocomplete` & `autofocus`

Day 14: Form Inputs — Selection Controls

Choice Inputs

- Checkboxes (`type="checkbox"`)
- Radio Buttons
 - `type="radio"`
 - Grouping by name
- checked Attribute

Dropdowns & Suggestions

- `<select>` Menu
 - `<option>`
 - `<optgroup>`
 - `selected`
 - `multiple`
- `<datalist>` Autocomplete

Day 15: Form Inputs — Date, File, Color & More

Specialized Inputs

- Date & Time
 - date
 - time
 - datetime-local
 - month
 - week
- color Picker
- file Upload
 - accept
 - multiple
- hidden Inputs

Multi-line & Buttons

- <textarea> (rows, cols)
- Buttons
 - type="submit"
 - type="reset"
 - type="button"
 - <button> element

Day 16: Form Validation & Grouping

Built-in Validation

- Constraint Attributes
 - required
 - pattern
 - min / max / step
- Validation State Preview (:valid / :invalid)
- novalidate Attribute
- Custom Validation Messages (concept)

Grouping & Structure

- <fieldset> & <legend>
- Accessible Grouping of Related Fields
- Logical Tab Order

Day 17: Semantic HTML & Document Outline

Why Semantics Matter

- Meaning vs Appearance
- Benefits: SEO, Accessibility, Maintainability
- "Div Soup" vs Semantic Markup

Structural Elements

- Page Landmarks
 - <header>
 - <nav>
 - <main>
 - <footer>
- Content Sections
 - <section>
 - <article>
 - <aside>
- Other Semantics
 - <details> & <summary>
 - <time>
 - <address>
- Document Outline Concept

Day 18: Multimedia — Audio, Video & Embeds

Native Media

- <audio>
 - controls
 - autoplay
 - loop
 - muted
- <video>
 - poster
 - preload
 - controls
- <source> for Multiple Formats
- <track> for Captions & Subtitles

Embedding External Content

- <iframe>
 - src
 - sandbox
 - allow
 - loading
- Embedding YouTube / Maps
- <embed> & <object>

Phase 2 • HTML — Accessibility, SEO & Practice

Day 19: Global Attributes & HTML Best Practices

Global Attributes

- Identity & Styling
 - id
 - class
 - style
 - title
- Behaviour
 - hidden
 - contenteditable
 - draggable
 - spellcheck
- tabindex & Focus Order
- Boolean Attributes

Custom Data

- data-* Custom Attributes
- Reading data-* in JS (concept)
- Use Cases for Data Attributes

Day 20: Accessibility (a11y) Fundamentals

Foundations

- Why Accessibility Matters
- Semantic HTML as the Base
- alt Text Best Practices
- Keyboard Navigation & Focus Management

ARIA & Standards

- ARIA Roles, States & Properties
- Common ARIA Attributes
 - aria-label
 - aria-labelledby
 - aria-describedby
 - aria-hidden
 - aria-live
- Skip Links
- WCAG & the POUR Principles

Day 21: SEO Fundamentals & Social Meta

On-Page SEO

- How Search Engines Read HTML
- Title & Meta Description for SEO
- Semantic Structure & Heading Hierarchy
- Descriptive, Crawlable Links

Sharing & Discovery

- Open Graph Tags
 - og:title
 - og:description
 - og:image
- Twitter Card Tags
- Canonical Links
- Indexing Controls
 - robots meta
 - sitemap.xml (concept)
- Structured Data / Schema.org (intro)

Day 22: HTML Validation & Project Day

Quality & Tooling

- Writing Clean, Maintainable HTML
- W3C HTML Validator
- Common HTML Mistakes
- Inspecting HTML with DevTools

Mini Project — Static Profile / Resume Page

- Semantic Page Structure
- Headings, Lists, Links & Images
- A Table & a Complete Form
- Embedded Media
- Validate the Final Markup

Phase 3 • CSS Fundamentals

Day 23: Introduction to CSS

CSS Basics

- What CSS Is & Its Purpose
- Syntax
 - Selector
 - Property
 - Value
 - Declaration Block
- Rulesets
- CSS Comments /* */

Adding CSS to HTML

- Three Methods
 - Inline (style attribute)
 - Internal (<style>)
 - External (<link>)
- Why External CSS Is Preferred
- Cascading: How Styles Combine

Day 24: Selectors — Basics

Core Selectors

- Universal Selector *
- Type / Element Selector
- Class Selector (.)
- ID Selector (#)

Combining & Choosing

- Grouping Selector (comma)

- Multiple Classes on One Element
- Class vs ID — When to Use Each

Day 25: Selectors — Combinators

Relationship Selectors

- Descendant Combinator (space)
- Child Combinator (>)
- Adjacent Sibling (+)
- General Sibling (~)

Practical Use

- Combining Multiple Selectors
- Selector Performance Basics
- Reading Complex Selectors

Day 26: Selectors — Attribute Selectors

Matching Attributes

- [attr]
- [attr="value"]
- [attr~="value"] (word match)
- [attr|= "value"] (prefix-lang)

Substring Matching

- [attr^="value"] (starts with)
- [attr\$="value"] (ends with)
- [attr*="value"] (contains)
- Case-Insensitive Flag (i)

Day 27: Pseudo-Classes

State & Interaction

- User-Action
 - :hover
 - :focus
 - :active
- Links
 - :link
 - :visited
- Form States
 - :checked
 - :disabled
 - :enabled
 - :required
 - :valid
 - :invalid

Structural Pseudo-Classes

- :first-child / :last-child
- :nth-child() & :nth-of-type()
- :first-of-type / :last-of-type / :only-child
- :not() • :is() • :where()
- :empty • :root • :target

Day 28: Pseudo-Elements

Generated Content

- ::before & ::after
- The content Property
- Decorative & Icon Use Cases

Text & UI Pseudo-Elements

- ::first-line & ::first-letter
- ::selection
- ::placeholder
- ::marker

Day 29: Specificity, Cascade & Inheritance

How Conflicts Resolve

- The Cascade Explained
- Specificity Calculation
 - Inline
 - ID
 - Class / attribute / pseudo-class
 - Element
- Source Order
- !important (and why to avoid it)

Inheritance

- Inherited vs Non-Inherited Properties
- Global Values
 - inherit
 - initial
 - unset
 - revert
- Avoiding Specificity Wars

Day 30: Colors in CSS

Color Notations

- Named Colors
- Hex
 - #RRGGBB
 - #RGB
 - 8-digit hex + alpha
- rgb() & rgba()
- hsl() & hsla()

Transparency & Modern Color

- Opacity vs Alpha Channel
- transparent & currentColor
- Modern: hwb(), lab(), lch() (overview)

Day 31: CSS Units

Absolute & Relative

- Absolute
 - px
 - pt, cm, in
- Font-Relative
 - em
 - rem
 - ch, ex
- Percentage (%)

Viewport & Choosing Units

- Viewport Units
 - vw / vh
 - vmin / vmax
- fr (grid fraction)
- rem vs em Deep Dive
- When to Use Which Unit

Day 32: The Box Model

Box Anatomy

- Four Layers
 - Content
 - Padding
 - Border
 - Margin
- Width & Height
- min-/max- width & height

Controlling the Box

- box-sizing
 - content-box
 - border-box
- The Universal border-box Reset
- Visualizing the Box Model in DevTools

Day 33: Margins & Padding

Spacing Properties

- Margin Shorthand & Longhand
- Padding Shorthand & Longhand
- Individual Sides (top/right/bottom/left)

Spacing Behaviour

- Margin Collapsing
- Negative Margins
- auto Margins for Centering
- Logical Properties (inline / block)

Day 34: Borders & Outlines

Borders

- Border Parts
 - border-width
 - border-style
 - border-color
- Border Shorthand & Individual Sides
- border-radius
 - Rounded corners
 - Circles, ellipses & pills

Outlines & Decoration

- outline vs border
- outline-offset
- border-image (overview)

Day 35: Backgrounds

Background Properties

- background-color
- background-image
- background-repeat
- background-position

Sizing & Layering

- background-size
 - cover
 - contain
 - explicit sizes
- background-attachment (fixed / scroll)
- background-origin & background-clip
- Multiple Backgrounds & Shorthand

Day 36: Gradients

Gradient Types

- linear-gradient()
- radial-gradient()
- conic-gradient()

Gradient Control

- Color Stops & Positions
- Repeating Gradients
- Layering Multiple Gradients
- Gradients as Backgrounds

Day 37: Typography — Fonts

Font Families

- font-family & Font Stacks
- Generic Families
 - serif
 - sans-serif
 - monospace
- Web-Safe Fonts

Custom & Web Fonts

- @font-face
- Google Fonts Integration
- font-display
- Variable Fonts (overview)
- font Shorthand

Day 38: Typography — Text Styling

Sizing & Spacing

- font-size, font-weight, font-style
- line-height
- letter-spacing & word-spacing
- text-indent

Alignment & Decoration

- text-align & text-justify
- text-decoration
- text-transform
- text-shadow
- Overflow & Wrapping
 - white-space
 - text-overflow: ellipsis
 - word-break / overflow-wrap

Day 39: The display Property

Display Values

- Outer Display
 - block
 - inline
 - inline-block
- none vs visibility: hidden
- Block vs Inline Differences & Use Cases

Modern Display

- display: flex (preview)
- display: grid (preview)
- display: contents

Day 40: Positioning

Position Values

- position: static
- position: relative
- position: absolute
- position: fixed
- position: sticky

Offsets & Stacking

- top / right / bottom / left
- Containing Block Concept
- z-index & Stacking Context
- Common Positioning Patterns

Phase 4 • CSS Layout — Flexbox, Grid & Patterns

Day 41: Float & Clear (Legacy Foundations)

Floats

- float: left / right / none
- Text Wrapping Around Floats
- Why Floats Were Used for Layout (history)

Containing Floats

- clear Property
- The Clearfix Technique
- Modern Alternatives (Flexbox / Grid)

Day 42: Overflow & Formatting Contexts

Overflow Control

- overflow
 - visible
 - hidden
 - scroll
 - auto
 - clip
- overflow-x & overflow-y
- Scroll Containers

Box Behaviour

- Block Formatting Context (BFC)
- Replaced vs Non-Replaced Elements
- Collapsing & Containment Basics

Day 43: Flexbox — Fundamentals

Flex Container

- display: flex
- Flex Container vs Flex Items
- Main Axis vs Cross Axis

Direction & Wrapping

- flex-direction
 - row / row-reverse
 - column / column-reverse
- flex-wrap
- flex-flow Shorthand

Day 44: Flexbox — Alignment

Aligning Items

- justify-content (main axis)
- align-items (cross axis)
- align-content (multi-line)
- align-self (per item)

Spacing & Centering

- gap in Flexbox
- Perfect Centering with Flexbox
- Distributing Space

Day 45: Flexbox — Item Sizing & Patterns

Flex Item Properties

- flex-grow
- flex-shrink
- flex-basis
- flex Shorthand
- order Property

Common Patterns

- Navigation Bars
- Card Rows
- Holy-Grail / Sidebar Layouts
- Equal-Height Columns

Day 46: CSS Grid — Fundamentals

Grid Container

- display: grid
- grid-template-columns & grid-template-rows
- The fr Unit
- repeat() Function

Tracks & Gaps

- Explicit vs Implicit Grid
- grid-auto-rows / grid-auto-columns
- gap (row-gap, column-gap)

Day 47: CSS Grid — Item Placement

Line-Based Placement

- Grid Lines & Numbering
- grid-column & grid-row (start / end)
- span Keyword

Area-Based Placement

- grid-area
- Named Grid Lines
- grid-template-areas
- Overlapping / Layering Items

Day 48: CSS Grid — Advanced

Flexible Tracks

- minmax() Function
- auto-fill vs auto-fit
- Responsive Grids Without Media Queries (RAM pattern)

Grid Alignment

- justify-items / align-items / place-items
- justify-self / align-self
- Subgrid (overview)
- Masonry-Style Layouts (overview)

Day 49: Flexbox vs Grid — Choosing Wisely

Decision Framework

- One-Dimensional vs Two-Dimensional
- Content-First vs Layout-First
- When Flexbox Wins / When Grid Wins

Combining Them

- Grid for Page, Flex for Components
- Nesting Flex Inside Grid
- Practice: Rebuild One Layout Both Ways

Day 50: Sizing & Intrinsic Layout

Intrinsic Sizing

- Intrinsic vs Extrinsic Sizing
- Keyword Sizes
 - min-content
 - max-content
 - fit-content
- aspect-ratio Property

Media Fitting

- object-fit (cover / contain)
- object-position
- Responsive Media Boxes

Day 51: CSS Functions

Math Functions

- calc()
- min() & max()
- clamp() for Fluid Sizing
- Nesting Functions

Practical Use

- Responsive Spacing with calc()
- Fluid Type with clamp()
- var() Preview

Day 52: CSS Custom Properties (Variables)

Declaring & Using

- Declaring --custom-properties
- var() Function & Fallback Values
- Scope: :root (global) vs local
- Inheritance of Custom Properties

Real-World Use

- Theming with Variables
- Updating Variables via JS (concept)
- Custom Properties vs Sass Variables

Day 53: Centering & Common Layout Patterns

Centering Toolkit

- Horizontal Centering (block / inline / flex / grid)
- Vertical Centering Techniques
- Dead-Center (both axes)

Reusable Patterns

- Sticky Footer
- Full-Bleed Sections
- Card & Media-Object Layouts
- Sidebar Layouts

Day 54: Layout Project Day

Build a Full Page Layout

- Header, Nav, Hero, Content, Sidebar, Footer
- Page Structure with Grid
- Components with Flexbox

Apply What You Learned

- Consistent Spacing System
- Box Model & Custom Properties
- Clean, Reusable Class Names

Phase 5 • Responsive & Adaptive Design

Day 55: Responsive Design Principles

Core Ideas

- What Responsive Web Design Is
- Fluid vs Fixed Layouts
- Mobile-First vs Desktop-First
- Responsive vs Adaptive Design

Foundations

- The Viewport Meta Tag (revisited)
- Breakpoints Philosophy
- Designing for Many Screen Sizes

Day 56: Media Queries — Fundamentals

Syntax & Types

- @media Rule Syntax
- Media Types
 - screen
 - print
 - all
- min-width & max-width Queries

Strategy

- Combining Conditions (and / or / not)
- Common Breakpoints
- Mobile-First Media Query Strategy

Day 57: Media Queries — Advanced Features

Feature Queries

- orientation (portrait / landscape)
- aspect-ratio
- resolution / device pixel ratio
- hover & pointer (coarse / fine)

User Preferences

- prefers-color-scheme (dark mode)
- prefers-reduced-motion
- Range Syntax (modern queries)

Day 58: Responsive Units & Fluid Typography

Flexible Sizing

- Relative Units in Responsive Design
- Viewport Units for Layout
- Responsive Spacing Systems

Fluid Type

- Fluid Typography with clamp()
- Scaling Strategies
- Avoiding Fixed Pixel Text

Day 59: Responsive Images & Media

Responsive Images

- srcset & sizes (revisited)
- <picture> for Art Direction
- Responsive Background Images

Media Fitting

- object-fit for Responsive Media
- Aspect-Ratio Boxes
- Lazy Loading for Performance

Day 60: Container Queries

Component-Level Responsiveness

- What Container Queries Are
- container-type & container-name
- @container Rule

Container Units & Comparison

- Container Query Units
 - cqw / cqh
 - cqi / cqj
- Container Queries vs Media Queries
- Designing Reusable Responsive Components

Day 61: Dark Mode & Theming

Building Themes

- prefers-color-scheme
- Light / Dark Theme Structure
- CSS Variables for Theming

Theme Switching

- color-scheme Property
- Theme Toggle (concept + CSS)
- Accessible Color Contrast

Day 62: Responsive Design Project Day

Build a Responsive Page

- Mobile, Tablet & Desktop Breakpoints
- Responsive Navigation (hamburger concept)
- Fluid Typography & Spacing

Validate

- Testing Across Viewports
- DevTools Device Mode
- Fixing Layout Breakpoints

Phase 6 • Visual Effects, Transforms & Animation

Day 63: Shadows & Depth

Shadow Properties

- box-shadow
 - offset-x / offset-y
 - blur
 - spread
 - color
- Inset Shadows
- Multiple Layered Shadows

Depth & Style

- text-shadow
- Elevation Systems
- Neumorphism (concept)

Day 64: Transforms — 2D

2D Transform Functions

- translate (X / Y)
- scale (X / Y)
- rotate
- skew

Controlling Transforms

- transform-origin
- Multiple / Combined Transforms
- Transforms vs Position for Movement

Day 65: Transforms — 3D

Adding Depth

- perspective
- rotateX / rotateY / rotateZ
- translate3d / translateZ

3D Scene Control

- transform-style: preserve-3d
- backface-visibility
- 3D Card Flip (concept)

Day 66: Transitions

Transition Properties

- transition-property
- transition-duration
- transition-timing-function
 - ease / linear
 - ease-in / ease-out
 - cubic-bezier()
- transition-delay

Using Transitions

- transition Shorthand
- Transitioning Multiple Properties
- What Can & Can't Be Transitioned

Day 67: Animations — Keyframes

Defining Animations

- @keyframes Rule
- from / to & Percentage Steps
- animation-name

Timing

- animation-duration
- animation-timing-function
- animation-delay

Day 68: Animations — Control

Playback Control

- animation-iteration-count
- animation-direction (normal / reverse / alternate)
- animation-fill-mode
- animation-play-state

Advanced Use

- animation Shorthand
- Chaining Multiple Animations
- steps() Timing Function

Day 69: Animation Performance & Best Practices

Performance

- What Triggers Repaints & Reflows
- Animating transform & opacity (GPU)
- will-change Property

Responsible Motion

- prefers-reduced-motion
- Targeting Smooth 60fps
- Avoiding Animation Jank

Day 70: Filters & Blend Modes

CSS Filters

- filter()
 - blur
 - brightness / contrast
 - grayscale / sepia
 - saturate / hue-rotate
 - drop-shadow / invert
- backdrop-filter (glassmorphism)

Blend Modes

- mix-blend-mode
- background-blend-mode
- Combining Filters Creatively

Day 71: Transitions & Animations — Practical

Interactive Effects

- Hover Effects
- Animated Buttons & Links
- Loading Spinners

Micro-Interactions

- Reveal / Fade-In Effects
- Scroll-Driven Animations (overview)
- Polish & Delight

Day 72: Clipping, Masking & Shapes

Clipping & Masking

- clip-path
 - polygon()
 - circle()
 - ellipse()
 - inset()
- mask Property (overview)

CSS Shapes

- shape-outside
- Wrapping Text Around Shapes
- Creative Layout Shapes

Phase 7 • Modern CSS, Architecture & Tooling

Day 73: CSS Architecture & Methodologies

Scaling CSS

- The Problem of Growing Stylesheets
- Naming Conventions
- File Organization Strategies

Methodologies

- BEM (Block — Element — Modifier)
- OOCSS Concept
- SMACSS Concept
- Utility-First (Tailwind philosophy)

Day 74: Modern CSS Selectors & Features

New Selectors

- :has() (the parent selector)
- :is() & :where() Deep Dive
- Specificity of Modern Selectors

Modern Capabilities

- Native CSS Nesting
- Cascade Layers (@layer)
- @scope (overview)
- Logical Properties (full)

Day 75: CSS Reset & Normalization

Why Reset

- Browser Default Styles
- Reset vs Normalize.css
- Inconsistencies Across Browsers

Modern Baseline

- A Modern CSS Reset (Andy Bell / Josh Comeau style)
- Building Your Own Reset
- Sensible Project Defaults

Day 76: CSS Preprocessors (Sass / SCSS)

Why & How

- What a Preprocessor Is & Why Use One
- Sass vs SCSS Syntax
- Compiling Sass to CSS

Core Features

- Variables & Nesting
- Partials & @use / @forward
- Mixins (@mixin / @include)
- Functions & Operators
- Inheritance (@extend)

Day 77: CSS Tooling & Workflow

Build Tooling

- PostCSS & Autoprefixer (concept)
- Vendor Prefixes
- CSS Minification

Quality & Workflow

- Linting with Stylelint
- Browser DevTools for CSS (deep dive)
- CSS-in-JS (overview)
- Build Tools Overview (Vite, etc.)

Day 78: Performance, Debugging & Best Practices

CSS Performance

- Selector & Paint Performance
- Critical CSS
- Reducing File Size

Maintainable CSS

- Debugging Layout Issues
- Common CSS Pitfalls
- Cross-Browser Compatibility
- Scalable, Maintainable Stylesheets

Phase 8 • Capstone Projects & Job-Ready Review

Day 79: Capstone Project 1 — Responsive Landing Page

Core Build

- Full Landing Page from Scratch
- Semantic HTML Structure
- Responsive Layout (Flexbox + Grid)
- Mobile-First Breakpoints

Polish & Quality

- Custom Properties & Theming
- Animations & Hover Effects
- Accessibility & HTML/CSS Validation
- Performance Pass

Sections to Include

- Hero with Call-to-Action
- Feature Cards Grid
- Testimonials / Pricing
- Contact Form & Footer

Day 80: Capstone Project 2 — Multi-Page Site & Final Review

Multi-Page Website

- Home, About & Contact Pages
- Consistent Design System
- Responsive Navigation
- Dark Mode Toggle

Reusable Components

- Buttons, Cards & Forms
- Shared Header & Footer
- BEM-Style Class Naming

Wrap-Up

- Review the Job-Ready Checklist
- Build a Portfolio of Projects
- Next Steps: JavaScript & a Framework

Final Job-Ready Checklist

If you can confidently build with and explain every item below, you have the HTML & CSS foundation expected of a junior front-end developer.

HTML Core

- ✓ Document Structure
- ✓ Semantic Elements
- ✓ Headings & Text
- ✓ Links & Anchors
- ✓ Images & Media
- ✓ Lists
- ✓ Tables
- ✓ Forms & Inputs
- ✓ Form Validation
- ✓ Global Attributes
- ✓ HTML Entities
- ✓ Iframes & Embeds

HTML Quality

- ✓ Accessibility (ARIA, alt, focus)
- ✓ WCAG Basics
- ✓ SEO & Meta Tags
- ✓ Open Graph / Social Meta
- ✓ Clean Markup
- ✓ HTML Validation
- ✓ DevTools Inspection

CSS Core

- ✓ Selectors (all types)
- ✓ Combinators
- ✓ Pseudo-Classes
- ✓ Pseudo-Elements
- ✓ Specificity & Cascade
- ✓ Inheritance
- ✓ Colors
- ✓ Units
- ✓ Box Model
- ✓ Spacing & Borders
- ✓ Backgrounds & Gradients
- ✓ Typography

CSS Layout

- ✓ display Property
- ✓ Positioning
- ✓ Float & Clear
- ✓ Overflow
- ✓ Flexbox
- ✓ CSS Grid
- ✓ Flexbox vs Grid
- ✓ Centering Patterns
- ✓ Intrinsic Sizing
- ✓ aspect-ratio
- ✓ Custom Properties
- ✓ CSS Functions

Responsive Design

- ✓ Mobile-First Approach
- ✓ Media Queries
- ✓ Advanced Media Features
- ✓ Fluid Typography
- ✓ Responsive Images
- ✓ Container Queries
- ✓ Dark Mode & Theming

Effects & Motion

- ✓ Shadows & Depth
- ✓ 2D Transforms
- ✓ 3D Transforms
- ✓ Transitions
- ✓ Keyframe Animations
- ✓ Animation Performance
- ✓ Filters & Blend Modes
- ✓ Clipping & Shapes

Architecture & Tooling

- ✓ BEM & Naming Conventions
- ✓ Modern Selectors (:has, :is)
- ✓ Cascade Layers & Nesting
- ✓ CSS Reset / Normalize
- ✓ Sass / SCSS Basics
- ✓ Autoprefixer & PostCSS
- ✓ Stylelint
- ✓ Performance & Debugging

Projects

- ✓ Responsive Landing Page
- ✓ Multi-Page Website
- ✓ Reusable Components
- ✓ Design System
- ✓ Accessible & Validated
- ✓ Portfolio-Ready Build